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tensorflow-riemopt: a library for optimization on Riemannian manifolds in TensorFlow

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Software

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Summary

tensorflow-riemopt is a TensorFlow (Abadi et al., 2016) library for training machine learning models whose parameters are constrained to Riemannian manifolds. It provides multiple manifolds with their Riemannian metrics, exponential and logarithmic maps, geodesics, retractions and vector transports, implemented either as closed-form expressions or approximations where applicable. The library also implements Riemannian versions of stochastic gradient descent (Bonnabel, 2013) and of the adaptive RMSprop (Roy et al., 2018) and Adam (Bécigneul & Ganea, 2019) algorithms. After a manifold is attached to a TensorFlow variable, Keras layers and training loops can be used without modification, resulting in a geometry-aware model that can be trained and served inside a TensorFlow Extended pipeline (Baylor et al., 2017).

The repository includes example implementations of SPDNet (Huang & Van Gool, 2017), GrNet (Huang et al., 2018), and LieNet (Huang et al., 2017) methods built on the library.

Statement of need

Several families of machine learning models keep their parameters on a non-Euclidean space. Hyperbolic embeddings represent data with latent hierarchies and are used in language modeling (Ganea et al., 2018; Nickel & Kiela, 2017, 2018; Sala et al., 2018), recommendation (Chamberlain et al., 2019) and image classification (Khrulkov et al., 2020). Weights on the Stiefel and Grassmann manifolds appear in networks for emotion and action recognition from video (Chakraborty & Vemuri, 2019; Huang et al., 2018; Huang & Van Gool, 2017) and in image set classification (Y. Li et al., 2015). Symmetric positive definite matrices represent diffusion tensors and covariance descriptors (Pennec et al., 2006), and the Lie groups $\mathrm{SO}(3)$ and $\mathrm{SE}(3)$ parameterize pose estimation and skeleton-based action recognition (Hou et al., 2018; Huang et al., 2017). Training of these models requires an optimizer whose updates are constrained to the manifold, which is not guaranteed by the conventional Euclidean optimizers.

State of the field

Riemannian stochastic optimization (Bécigneul & Ganea, 2019; Bonnabel, 2013; Kasai et al., 2019; Roy et al., 2018) has support in PyTorch through Geoopt (Kochurov et al., 2020) and McTorch (Meghwanshi et al., 2018). TensorFlow, and the related TFX and TensorFlow Serving frameworks, had no maintained equivalent. tensorflow-riemopt closes this gap by providing manifold-constrained variables and Riemannian optimizers that replace their Euclidean counterparts in any existing TensorFlow / Keras pipeline. It also exposes a low-level API for the geometric operations, so new manifolds and algorithms are implemented and tested against the existing alternatives.

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Across related work, Geoopt (Kochurov et al., 2020) and McTorch (Meghwanshi et al., 2018) implement Riemannian stochastic optimizers for PyTorch, while Pymanopt (Townsend et al., 2016) implements deterministic solvers and differentiates cost functions supported by TensorFlow, PyTorch, and JAX backends. However, Pymanopt computes the manifold operations in NumPy and does not provide stochastic optimizers, so it does not support minibatch training of neural networks. Geomstats (Miolane et al., 2020) implements additional manifolds and differential-geometric constructions, providing a scikit-learn API (Pedregosa et al., 2011) with NumPy, PyTorch, and Autograd backends. Manopt (Boumal et al., 2014), Manifolds.jl (Axen et al., 2023) and Rieoptax (Utpala et al., 2022) provide alternative implementations in MATLAB, Julia, and JAX correspondingly.

Software design

In tensorflow-riemopt, the manifold constraint is a property of the variable, that can be set by `assign_to_manifold(variable, manifold)` function. Therefore, a model can combine manifold-constrained and unconstrained layers, which are treated accordingly by the Riemannian optimizers. After training, the exported model contains only standard TensorFlow operations, so TensorFlow TFX components do not depend on the library.

```
import tensorflow as tf
from tensorflow_riemopt.variable import assign_to_manifold
from tensorflow_riemopt.manifolds import StiefelEuclidean
from tensorflow_riemopt.optimizers import RiemannianSGD


class BiMap(tf.keras.layers.Layer):
    """Bilinear mapping layer of SPDNet with a weight on the Stiefel manifold."""

    def __init__(self, output_dim, **kwargs):
        super().__init__(**kwargs)
        self.output_dim = output_dim

    def build(self, input_shape):
        self.w = self.add_weight(name="w", shape=[input_shape[-1], self.output_dim])
        assign_to_manifold(self.w, StiefelEuclidean())

    def call(self, inputs):
        return tf.transpose(self.w) @ inputs @ self.w


model = tf.keras.Sequential([BiMap(32), tf.keras.layers.Flatten(), tf.keras.layers.Dense
model.compile(optimizer=RiemannianSGD(learning_rate=0.01), loss="sparse_categorical_cros
```

As in Manopt (Boumal et al., 2014), each manifold subclasses `Manifold` and implements the methods the optimization algorithms call: the Riemannian metric `inner(x, u, v)`, the tangent-space projection `proju(x, u)`, a retraction `retr(x, u)` and a vector transport `transp(x, y, v)`. Exponential and logarithmic maps and parallel transport are additional methods that can be added for adaptive optimization support.

```
from tensorflow_riemopt.manifolds import Sphere

S = Sphere()
x = S.projx(tf.constant([0.1, -0.1, 0.1]))
u = S.proju(x, tf.constant([1.0, 1.0, 1.0]))
v = S.proju(x, tf.constant([-0.7, -1.4, 1.4]))
y = S.exp(x, v)
```

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```
u_y = S.transp(x, y, u)
v_y = S.transp(x, y, v)
```

Geometric operations on the sphere $\mathbb{S}^2$. The exponential map takes x along v to y, and vector transport carries u and v from $T_x\mathcal{M}$ to $T_y\mathcal{M}$.

Figure 1: Geometric operations on the sphere $\mathbb{S}^2$. The exponential map takes x along v to y, and vector transport carries u and v from $T_x\mathcal{M}$ to $T_y\mathcal{M}$.

The library implements the exponential and logarithmic map in closed form where it is known. Otherwise, it falls back to a published approximate algorithm, e.g., the matrix-algebraic logarithm on the Stiefel manifold under the canonical metric (Zimmermann, 2017) and the Cayley retraction (J. Li et al., 2020). Every manifold method is composed of TensorFlow operations, so automatic differentiation, graph execution and eager execution do not require manifold-specific code.

The optimizers implement the Keras optimizer interface, including the sparse update path for `tf.IndexedSlices`, so in an embedding table on the Poincaré ball or the hyperboloid only the rows present in a batch are updated. The `layers` module provides such a table as `ManifoldEmbedding`. Optimizer state and hyperparameters are serialized through the Keras configuration mechanism, so a compiled model is saved and restored with the standard calls.

Research impact statement

As of 30 October 2026, Google Scholar lists 12 works citing the preprint (Smirnov, 2021), the GitHub repository has ~100 stars and ~20 forks. ANTHEM (Choudhary et al., 2022), a hyperbolic entity model for product search, imports the library in the public implementation released by Amazon Science, which installs it from our repository under its former name tensorflow-manopt. The monograph on Riemannian geometry and geometric statistics by Guigui et al. (2023) and the survey of geometric optimization for deep learning by Fei et al. (2025) cite the library, and Axen et al. (2023) include it in their cross-library benchmark of manifold operations. The citing research includes Riemannian Hamiltonian methods for min-max optimization (Han et al., 2023), drug-target interaction prediction (Binatlı & Gönen, 2023) and differentially private Fréchet means on symmetric positive definite matrices (Utpala et al., 2023).

The package is distributed on PyPI, documented on Read the Docs, and tested in continuous integration with coverage reporting against current TensorFlow releases.

AI usage disclosure

A generative AI tool was used to prepare the manuscript draft from an earlier preprint describing the library (Smirnov, 2021). The final draft was reviewed and edited to verify its claims against the repository and the cited works.

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